DEEP LEARNING APPLICATIONS

Mini Project – CSCI – 5501

Presented By : Siddik A. Patel

B00986974

INTRODUCTION & MOTIVATION

ResNet18 provides a strong baseline for image recognition, and adding Squeeze-and-Excitation (SE) blocks further enhances feature recalibration. However, the common use of ReLU can limit expressiveness by discarding negative activations.

In this project, I replace ReLU with the Gated Linear Unit (GLU) to leverage its gating mechanism for richer feature interactions, while using dropout to improve generalization. The goal is to test whether these modifications can boost ResNet18's accuracy and robustness without adding significant complexity.

PROBLEM OF STUDY

Can modified channel attention mechanisms improve the classification accuracy of ResNet-18 on malaria cell detection tasks compared to the standard ResNet-18 architecture?

- This project proposes a ResNet-18 with modified channel attention, replacing ReLU with GLU (Gated Linear Units) for more effective feature gating.
- It evaluates whether this architecture improves classification accuracy and generalization over plain ResNet-18.

RELATED WORK

- The study conducted by K. Hemachandran et. el. compared CNN, MobileNetV2, and ResNet50 to identify infected cells using the same dataset. The results indicate that MobileNetV2 achieved the highest accuracy at 97.06% [1].

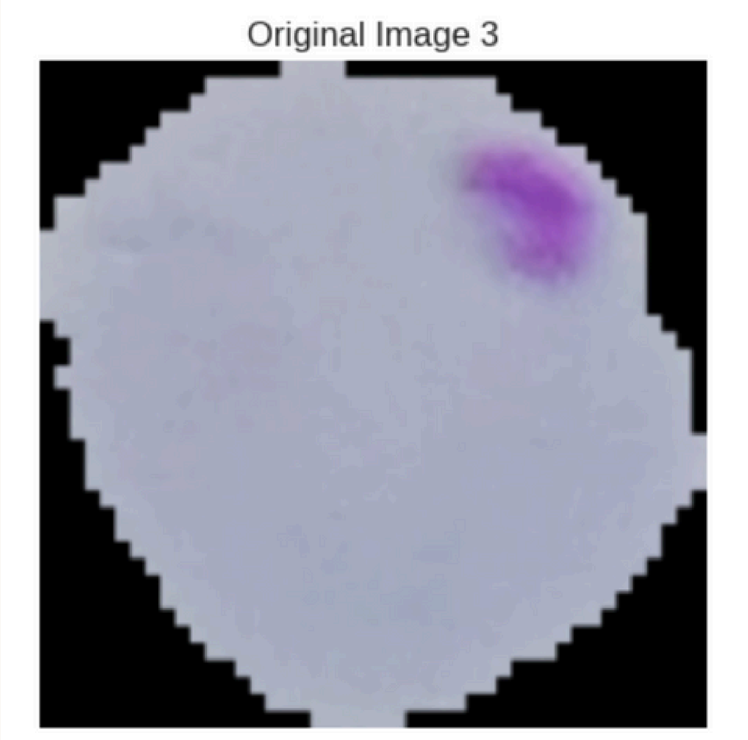
- SENet (CVPR 2018) was the first to model channel-wise attention explicitly, moving beyond earlier spatial-only methods and cutting ResNet-50 Top-5 ImageNet error from 7.48% to 6.62%, winning ILSVRC 2017 [2], and shows 93% accuracy for fracture image classification [3].

- The EDRI model (hybrid of EfficientNetB2, DenseNet, ResNet, and Inception) achieved 97.68% accuracy in malaria detection by capturing diverse, multi-scale features and ensuring robustness across varied imaging conditions [4].

METHODOLOGY

Dataset and Preprocessing

- Dataset: 27,558 Malaria cell images from Kaggle ("cell-images-for-detecting-malaria") [5]
- Classes: Binary classification (Uninfected vs. Parasitized)
- Data Split: 70% training, 15% validation, 15% testing
- Preprocessing:
  - Resize to 224×224 pixels
  - Data augmentation for training (horizontal flip, rotation)
  - Image normalization



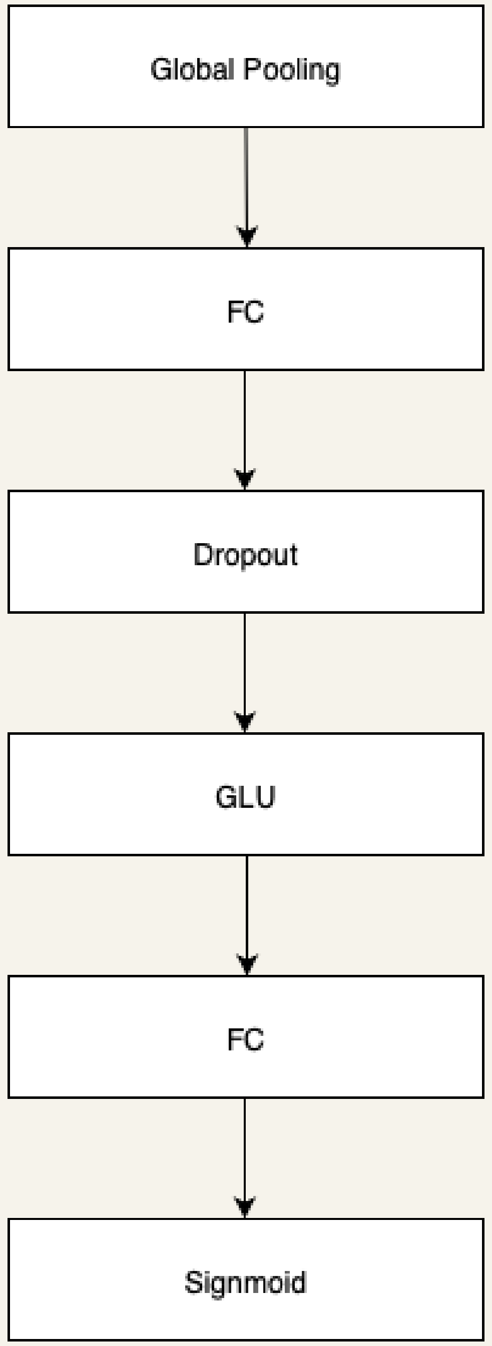
METHODOLOGY (CONTD...)

Key Features:

- Reduction ratio of 16 for computational efficiency
- Dropout (0.1) for regularization
- GLU (Gated Linear Unit) activation function instead of ReLU
- Sigmoid activation for final attention weights

ResNet with Attention Integration:

- Integrates channel attention modules after each ResNet block's second convolution
- Maintains residual connections
- Applied to all layers (layer1, layer2, layer3, layer4)



METHODOLOGY (CONTD...)

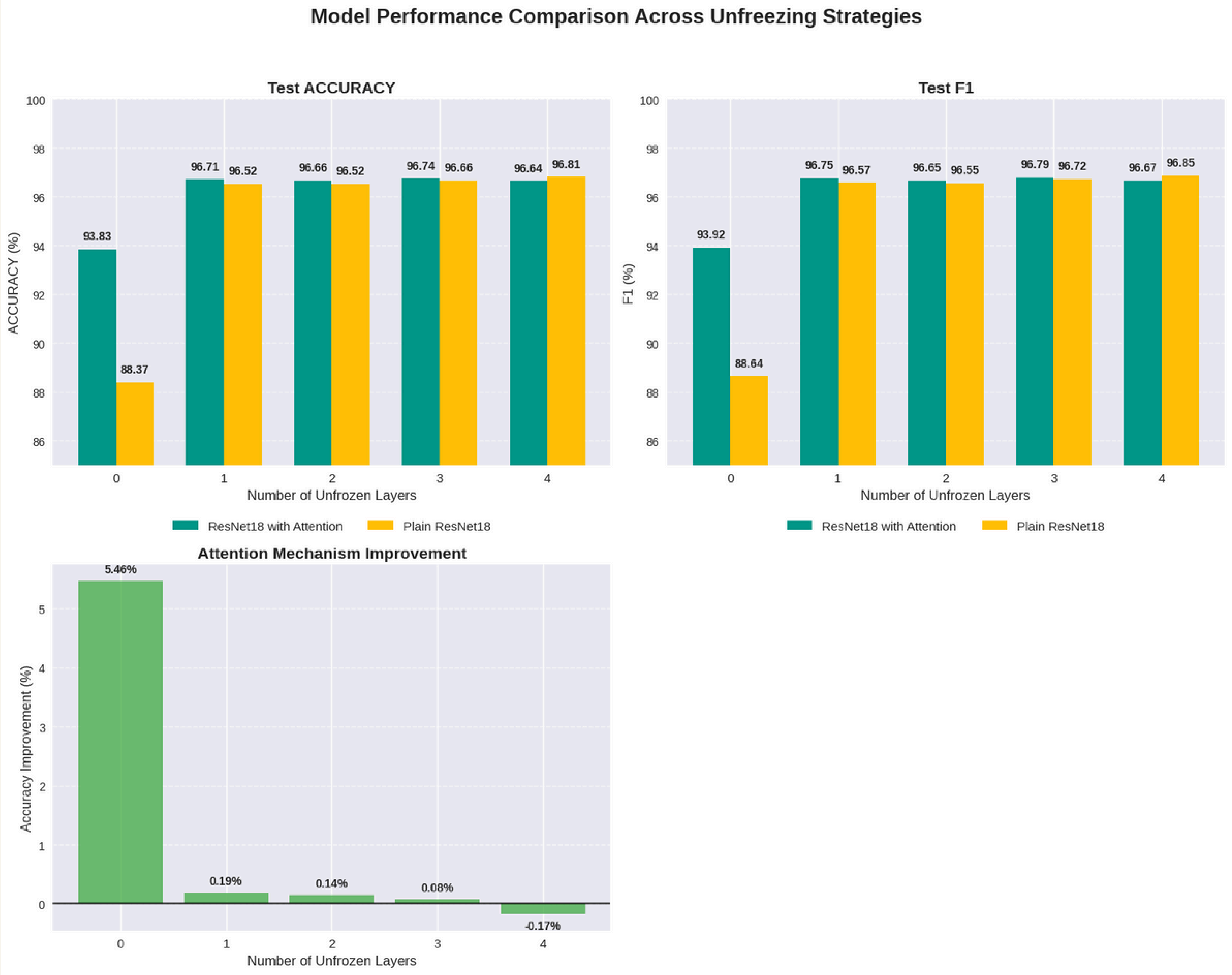
The study systematically evaluates 5 different configurations:

- 0 layers unfrozen: Only final classification layer and attention modules trainable
- 1 layer unfrozen: layer4 + classification layer + attention modules
- 2 layers unfrozen: layer3, layer4 + classification layer + attention modules
- 3 layers unfrozen: layer2, layer3, layer4 + classification layer + attention modules
- 4 layers unfrozen: All layers trainable

Training Configuration

- Optimizer: Adam with learning rate 1e-4
- Loss Function: CrossEntropyLoss
- Epochs: 3 per configuration
- Batch Size: 64 (due to compute constraints)
- Evaluation Metrics: Accuracy, F1-score, Loss

EXPERIMENTS AND RESULTS



DISCUSSION/LIMITATIONS

- **Limited Training Duration:** Only 3 epochs per configuration may not capture full model potential
- **Modest Improvements:** 0.16% accuracy gain, while positive, may not be statistically significant
- **Architecture Constraints:** Channel attention only, spatial attention not explored
- **Hyperparameter Optimization:** Limited exploration of attention module hyperparameters (reduction ratio, dropout)

CONCLUSION

This study shows that adding channel attention makes ResNet-18 noticeably better at classifying malaria cells. The best results came from gradually unfreezing one to four layers during training, which struck a good balance between improving accuracy and keeping the model stable. If the computation resources are limited, keeping all the layers frozen helps save time while producing comparatively better results than plain ResNet models.

FUTURE WORK

- **Extended Training:** Increase epoch count and implement early stopping for more robust results
- **Hyperparameter Optimization:** Systematic exploration of reduction ratios, dropout rates, and attention placement
- **Attention Mechanism Comparison:** Compare channel attention with spatial attention, self-attention, and hybrid approaches

REFERENCES

[1] K. Hemachandran et el., “Performance Analysis of Deep Learning Algorithms in Diagnosis of Malaria Disease,” National Institutes of Health, Link: Performance Analysis of Deep Learning Algorithms in Diagnosis of Malaria Disease [Feb. 2023]

[2] J. Hu, L. Shen, and G. Sun, “Squeeze-and-Excitation Networks,” 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition,” Link: https://ieeexplore.ieee.org/document/8578843 [Jun. 2018].

[3] Hua Wang et el., “Harnessing ResNet50 and SENet for enhanced ankle fracture identification,” National Institutes of Health, Link: Harnessing ResNet50 and SENet for enhanced ankle fracture identification [Apr. 2024].

[4] Sorio Boit, Rajvardhan Patil, “An Efficient Deep Learning Approach for Malaria Parasite Detection in Microscopic Images,” MDPI, Link: An Efficient Deep Learning Approach for Malaria Parasite Detection in Microscopic Images [Dec. 2024]

[5] Arunava, “Malaria Cell Images Dataset,” Kaggle, Link: https://www.kaggle.com/datasets/iarunava/cell-images-for-detecting-malaria [2018]

THANK YOU

Presented By : Siddik A. Patel

B00986974